

Computer Networks Overview

The OSI Model

The OSI model describes network communication in seven layers: Physical, Data Link, Network, Transport, Session, Presentation, and Application. Each layer provides services to the layer above it and relies on services from the layer below it.

TCP/IP Protocol Suite

The Transmission Control Protocol (TCP) provides reliable, ordered, and error-checked delivery of data between applications, using a three-way handshake (SYN, SYN-ACK, ACK) to establish connections. The Internet Protocol (IP) handles addressing and routing of packets across networks.

User Datagram Protocol (UDP) is a connectionless protocol that offers faster but unreliable transmission, commonly used in applications like video streaming and online gaming where speed matters more than guaranteed delivery.

Routing

Routers forward packets between networks based on routing tables, which can be built using protocols such as OSPF (Open Shortest Path First) and BGP (Border Gateway Protocol).

DNS

The Domain Name System (DNS) translates human-readable domain names, such as example.com, into IP addresses that computers use to identify each other on the network.

Network Security

Firewalls filter incoming and outgoing traffic based on predefined security rules. A Virtual Private Network (VPN) encrypts traffic between a client and a server, protecting data from interception on untrusted networks.

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